## Introduction

A Hotel Management System (HMS) is a database-driven solution used to manage hotel operations such as reservations, rooms, customer details, billing, and services.

## Objectives

To manage hotel room bookings
To maintain customer records
To handle billing
To manage staff
To ensure data accuracy
To provide reporting

## Entities and Attributes

Customer: Customer_ID, Name, Gender, Address, Contact, Email. Room: Room_ID, Room_Number, Room_Type, Price. Reservation: Reservation_ID, Customer_ID, Room_ID.

## ER Diagram

CUSTOMER 1 ——∞ RESERVATION ∞——1 ROOM. CUSTOMER 1 ——∞ SERVICE_USAGE ∞——1 SERVICE. RESERVATION 1 ——1 BILLING.

## Table Structure

Customer Table: Customer_ID INT PK, Name VARCHAR. Room Table: Room_ID INT PK, Room_Number INT.

## Functional Requirements

Customer Management, Room Management, Billing, Service Management, Staff Management, Security

## DBMS Features

ACID Properties, Integrity Constraints, Recovery, Scalability

## Advantages

Faster operations, Improved satisfaction, Secure data

## Challenges

Scalability, Integration, Cybersecurity

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